

```
In [18]: import pandas as pd
df = pd.read_csv('C:/Users/RUDRAKSHI/Downloads/train.csv/train.csv', encoding='lati
```

```
In [19]: df.head()
```

```
Out[19]:
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	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States

```
In [20]: df.info()
df.isnull().sum()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                9800 non-null  int64
1   Order ID              9800 non-null  object
2   Order Date            9800 non-null  object
3   Ship Date             9800 non-null  object
4   Ship Mode             9800 non-null  object
5   Customer ID           9800 non-null  object
6   Customer Name         9800 non-null  object
7   Segment              9800 non-null  object
8   Country               9800 non-null  object
9   City                 9800 non-null  object
10  State                9800 non-null  object
11  Postal Code          9789 non-null  float64
12  Region               9800 non-null  object
13  Product ID           9800 non-null  object
14  Category             9800 non-null  object
15  Sub-Category         9800 non-null  object
16  Product Name         9800 non-null  object
17  Sales                9800 non-null  float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB

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Out[20]: Row ID                0
         Order ID             0
         Order Date           0
         Ship Date            0
         Ship Mode            0
         Customer ID          0
         Customer Name        0
         Segment             0
         Country              0
         City                 0
         State                0
         Postal Code          11
         Region               0
         Product ID           0
         Category             0
         Sub-Category         0
         Product Name         0
         Sales                0
         dtype: int64

```

```
In [22]: df['Order Date'].head(10)
```

```
Out[22]: 0    08/11/2017
         1    08/11/2017
         2    12/06/2017
         3    11/10/2016
         4    11/10/2016
         5    09/06/2015
         6    09/06/2015
         7    09/06/2015
         8    09/06/2015
         9    09/06/2015
        Name: Order Date, dtype: object
```

```
In [23]: df = df.drop_duplicates()
```

```
In [25]: print(df.columns.tolist())
```

```
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales']
```

```
In [26]: print(df['Sales'].sum())
```

```
2261536.7827000003
```

```
In [27]: df.groupby('Category')['Sales'].sum().sort_values(ascending=False)
df.groupby('Region')['Sales'].sum().sort_values(ascending=False)
df.groupby('Sub-Category')['Sales'].sum().sort_values(ascending=False)
```

```
Out[27]: Sub-Category
Phones          327782.4480
Chairs          322822.7310
Storage         219343.3920
Tables          202810.6280
Binders         200028.7850
Machines        189238.6310
Accessories     164186.7000
Copiers         146248.0940
Bookcases       113813.1987
Appliances      104618.4030
Furnishings     89212.0180
Paper           76828.3040
Supplies        46420.3080
Art             26705.4100
Envelopes       16128.0460
Labels          12347.7260
Fasteners       3001.9600
Name: Sales, dtype: float64
```

```
In [29]: df['Order Date'] = pd.to_datetime(df['Order Date'], format='mixed')
```

```
In [30]: print(df['Order Date'].dtype)
```

```
datetime64[ns]
```

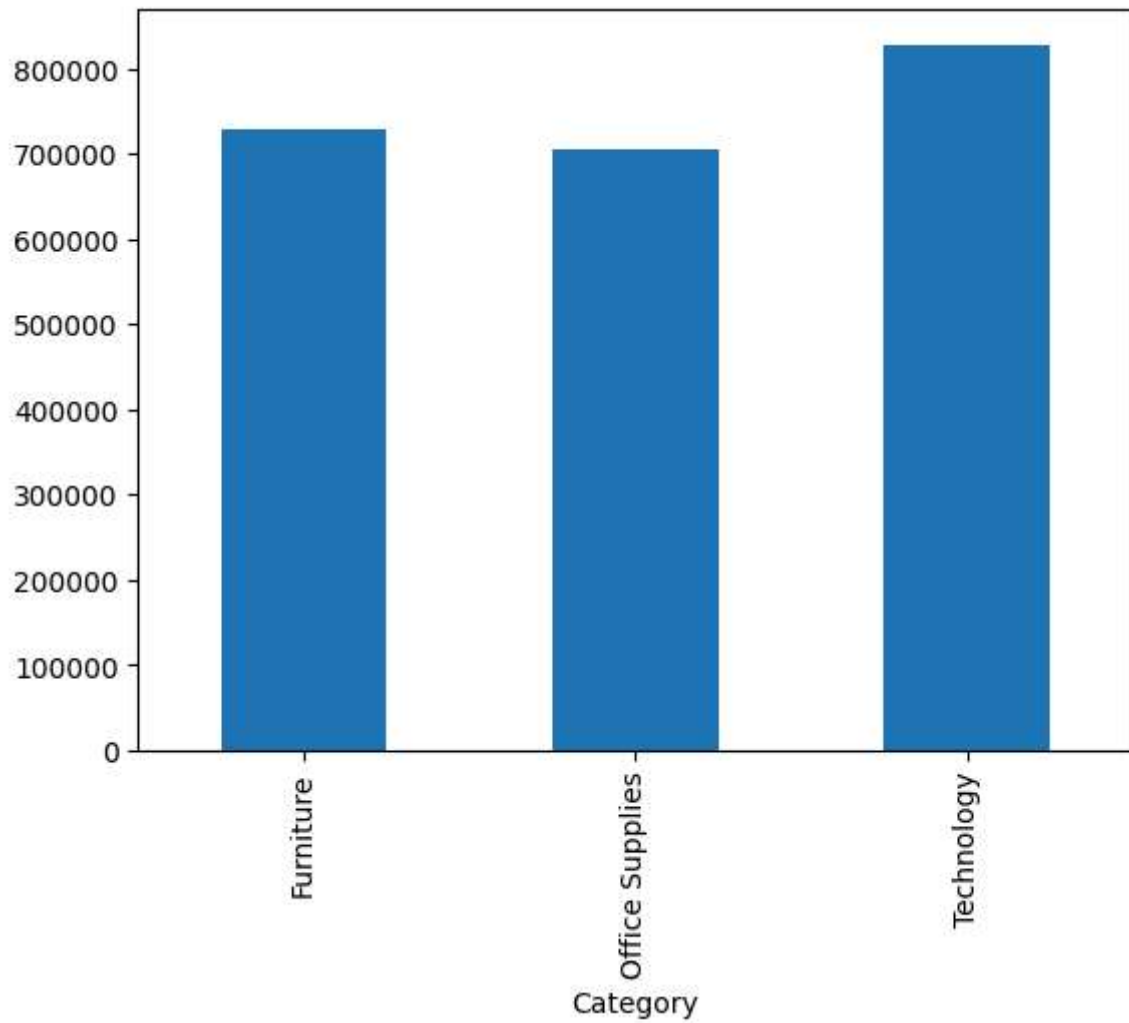
```
In [31]: df.groupby(df['Order Date'].dt.month)['Sales'].sum()
```

```
Out[31]: Order Date
1      155990.9154
2      131153.2594
3      212256.1344
4      142984.2481
5      166000.7467
6      142034.7713
7      159472.5865
8      207601.7939
9      245155.0671
10     183851.9562
11     268768.7885
12     246266.5152
Name: Sales, dtype: float64
```

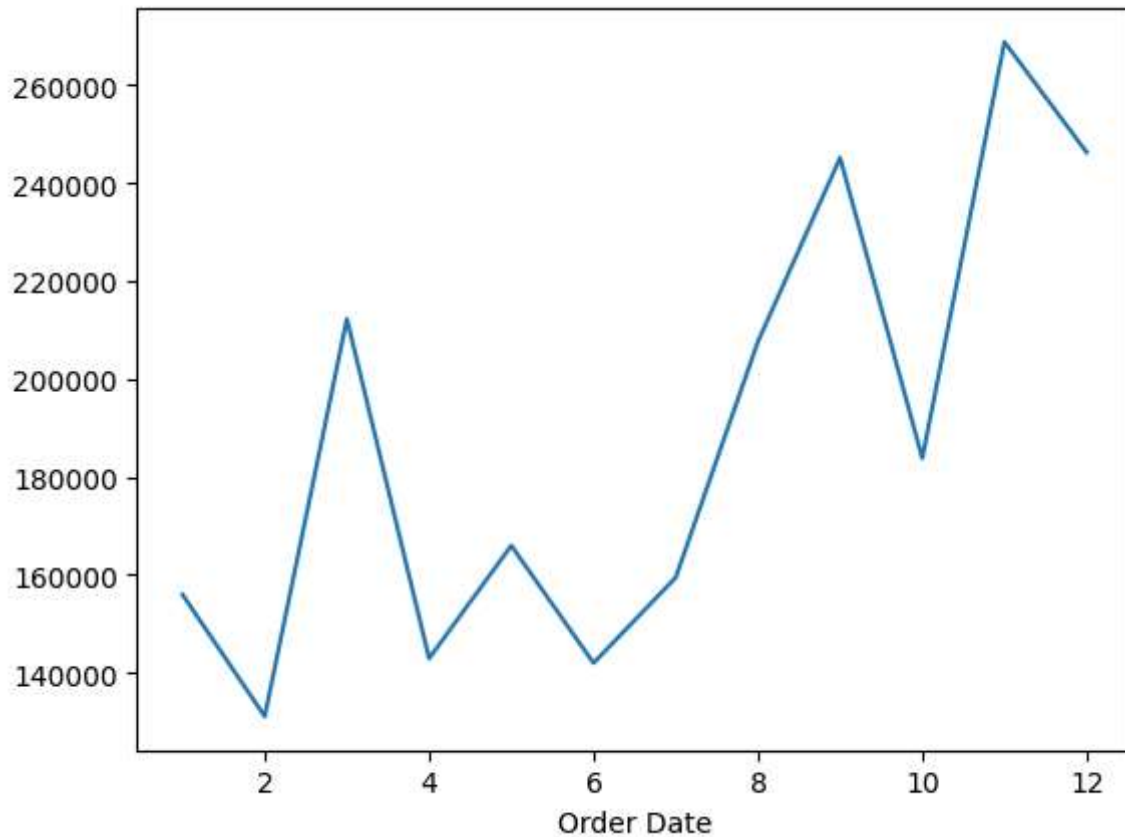
```
In [32]: df.groupby('Customer Name')['Sales'].sum().sort_values(ascending=False).head(10)
```

```
Out[32]: Customer Name
Sean Miller      25043.050
Tamara Chand     19052.218
Raymond Buch     15117.339
Tom Ashbrook     14595.620
Adrian Barton    14473.571
Ken Lonsdale     14175.229
Sanjit Chand     14142.334
Hunter Lopez     12873.298
Sanjit Engle     12209.438
Christopher Conant 12129.072
Name: Sales, dtype: float64
```

```
In [33]: import matplotlib.pyplot as plt
df.groupby('Category')['Sales'].sum().plot(kind='bar')
plt.show()
```



```
In [34]: df.groupby(df['Order Date'].dt.month)['Sales'].sum().plot(kind='line')
plt.show()
```



Based on your actual outputs, here are 3 solid, numbers-backed insights: Insight 1 — Technology drives the most revenue From your bar chart (groupby('Category')):

Technology is the top-performing category, generating roughly 830,000 in sales— ahead of Furniture (730,000) and Office Supplies (~700,000). At the sub – category level, Phones (327,782) and Chairs (\$322,822) are the two highest revenue generators individually. This suggests the business should prioritize inventory and marketing investment in Technology and Furniture's top sub-categories, since they're carrying a disproportionate share of revenue.

Insight 2 — Sales are highly seasonal, peaking at year-end From your line chart (groupby(Order Date.dt.month)):

Sales show a strong seasonal pattern, peaking sharply in November (268,768) and December (246,266) — likely driven by holiday shopping and year-end business purchasing. In contrast, February is the weakest month (\$131,153), less than half of November's total. This means the business should plan inventory stock-up and marketing campaigns ahead of Q4, while considering promotions in Q1 (especially February) to smooth out the slow period.

Insight 3 — Revenue is concentrated in a small group of top customers From your groupby('Customer Name') output:

The top customer, Sean Miller, alone contributes 25,043 in sales, with the top 10 customers (including Tamara Chandat 19,052 and Raymond Buch at \$15,117) together driving a significant chunk of total revenue. This concentration means the business is somewhat dependent on a small number of high-value customers — losing even one or two of them could noticeably impact revenue, so a customer retention/loyalty strategy for this segment would reduce that risk.